

RCV Data Missingness Audit

```
library(dplyr)
library(tidyr)
library(readr)
```

Overview

This document audits how many rows are dropped or have missing values at each stage of the cleaning pipeline. The goal is to identify where data is lost.

Alaska 2020

```
ak_results <- read.csv("../data/raw/alaska_all_results_2020.txt", header = FALSE)
names(ak_results) <- c("district", "race", "V3", "V4", "choice", "party", "V7", "votes", "V9")

ak_results <- ak_results %>%
  select(district, race, choice, party, votes) %>%
  filter(!choice %in% c("Number of Precincts for Race",
                       "Number of Precincts Reporting", "Registered Voters",
                       "Times Counted"))

ak_pres <- ak_results %>% filter(race == "U.S. President / Vice President") %>%
  select(district, race, party, votes) %>%
  pivot_wider(names_from = party, values_from = votes)

ak_m2 <- ak_results %>% filter(race == "Ballot Measure No. 2 - 19AKBE") %>%
  pivot_wider(names_from = choice, values_from = votes) %>%
  select(district, race, YES, NO)

cat("Pres precincts:", nrow(ak_pres), "\n")
```

Pres precincts: 580

```
cat("Ballot measure precincts:", nrow(ak_m2), "\n")
```

Ballot measure precincts: 579

```
# right_join keeps all pres, drops m2-only
ak_merged <- right_join(ak_m2, ak_pres, by = "district")
cat("After right_join:", nrow(ak_merged), "\n")
```

After right_join: 580

```
# What's lost?  
cat("M2 precincts not in pres:", sum(!ak_m2$district %in% ak_pres$district), "\n")
```

M2 precincts not in pres: 0

```
cat("Pres precincts not in M2:", sum(!ak_pres$district %in% ak_m2$district), "\n")
```

Pres precincts not in M2: 1

```
ak_merged <- ak_merged %>%  
  select(-race.x, -race.y) %>%  
  mutate(yes_share = YES / (YES + NO),  
         biden_share = DEM / (DEM + CON + ALI + GRN + LIB + NOM + REP))  
  
cat("NaN in yes_share:", sum(is.nan(ak_merged$yes_share)), "\n")
```

NaN in yes_share: 1

```
cat("NaN in biden_share:", sum(is.nan(ak_merged$biden_share)), "\n")
```

NaN in biden_share: 1

```
cat("NA in yes_share:", sum(is.na(ak_merged$yes_share)), "\n")
```

NA in yes_share: 2

```
cat("NA in biden_share:", sum(is.na(ak_merged$biden_share)), "\n")
```

NA in biden_share: 1

Massachusetts 2020

```
mass_rcv <- read.csv("../data/raw/massachusetts_rcv_2020.csv")  
mass_pres <- read.csv("../data/raw/massachusetts_pres_2020.csv")  
  
mass_pres <- mass_pres %>%  
  filter(!is.na(No.Preference), City.Town != "TOTALS") %>%  
  rename(Locality = City.Town) %>%  
  mutate(precinct = paste(Locality, Ward, Pct, sep = "_")) %>%  
  rename(biden = Joseph.R..Biden..Jr., total = Total.Votes.Cast) %>%  
  mutate(biden = parse_number(biden), total = parse_number(total)) %>%  
  mutate(biden_share = biden / total)  
  
mass_rcv <- mass_rcv %>%  
  filter(Locality != "TOTALS") %>%
```

```
mutate(precinct = paste(Locality, Ward, Pct, sep = "_")) %>%
mutate(Yes = parse_number(Yes), total = parse_number(Total.Votes.Cast)) %>%
mutate(yes_share = Yes / total)

cat("Pres precincts:", n_distinct(mass_pres$precinct), "\n")
```

Pres precincts: 2173

```
cat("RCV precincts:", n_distinct(mass_rcv$precinct), "\n")
```

RCV precincts: 2173

```
# What does inner_join lose?
inner <- inner_join(mass_pres, mass_rcv, by = "precinct")
full <- full_join(mass_pres, mass_rcv, by = "precinct")
cat("inner_join rows:", nrow(inner), "\n")
```

inner_join rows: 2173

```
cat("full_join rows:", nrow(full), "\n")
```

full_join rows: 2173

```
cat("Rows DROPPED by inner_join:", nrow(full) - nrow(inner), "\n")
```

Rows DROPPED by inner_join: 0

```
pres_only <- anti_join(mass_pres, mass_rcv, by = "precinct")
rcv_only <- anti_join(mass_rcv, mass_pres, by = "precinct")
cat("Precincts in pres only:", nrow(pres_only), "\n")
```

Precincts in pres only: 0

```
cat("Precincts in RCV only:", nrow(rcv_only), "\n")
```

Precincts in RCV only: 0

```
if (nrow(pres_only) > 0) {
  cat("Sample pres-only precincts:\n")
  print(head(pres_only$precinct, 10))
}
if (nrow(rcv_only) > 0) {
  cat("Sample RCV-only precincts:\n")
  print(head(rcv_only$precinct, 10))
}
```

Maine 2016

```

maine_pres <- read.csv("../data/raw/maine_pres_2016.csv", fileEncoding = "UTF-8-BOM")
maine_ref <- read.csv("../data/raw/maine_rcv_2016.csv", fileEncoding = "UTF-8-BOM")

# Use positional indexing (col names vary with BOM)
names(maine_ref)[c(1, 3, 16, 17, 18)] <- c("county", "dist", "rcv_yes", "rcv_no", "rcv_blank")
maine_ref <- maine_ref %>%
  filter(county != "", county != "CTY") %>%
  select(county, dist, rcv_yes, rcv_no, rcv_blank) %>%
  filter(dist != "MUNICIPALITY") %>%
  mutate(across(rcv_yes:rcv_blank, parse_number)) %>%
  mutate(rcv_share = rcv_yes / (rcv_yes + rcv_no))

names(maine_pres)[1:2] <- c("county", "dist")
maine_pres <- maine_pres %>%
  filter(county != "", county != "CTY") %>%
  select(county, dist, clinton = Clinton..Hillary.R., total = TBC) %>%
  filter(dist != "Town") %>%
  mutate(across(clinton:total, parse_number)) %>%
  mutate(clinton_share = clinton / total)

cat("Pres districts:", n_distinct(maine_pres$dist), "\n")

```

Pres districts: 532

```
cat("Ref districts:", n_distinct(maine_ref$dist), "\n")
```

Ref districts: 531

```

maine_full <- full_join(maine_pres, maine_ref, by = "dist")

pres_only <- anti_join(maine_pres, maine_ref, by = "dist")
ref_only <- anti_join(maine_ref, maine_pres, by = "dist")
cat("Districts in pres only:", nrow(pres_only), "\n")

```

Districts in pres only: 39

```
cat("Districts in ref only:", nrow(ref_only), "\n")
```

Districts in ref only: 38

```
cat("Total rows after full_join:", nrow(maine_full), "\n")
```

Total rows after full_join: 570

```
cat("NA in clinton_share:", sum(is.na(maine_full$clinton_share)), "\n")
```

NA in clinton_share: 38

```
cat("NA in rcv_share:", sum(is.na(maine_full$rcv_share)), "\n")
```

NA in rcv_share: 39

```
if (nrow(pres_only) > 0) {
  cat("\nSample pres-only districts:\n")
  print(head(pres_only$dist, 10))
}
```

Sample pres-only districts:

```
[1] "BIG TWENTY TWP/T15 R15 WELS"
[2] "E TWP"
[3] "MOLUNKUS TWP"
[4] "T11 R4 WELS TWP"
[5] "T12/R13 WELS/T9 R8 WELS"
[6] "T15 R6 TWP"
[7] "T17 R3 VAN BUREN COVE TWP"
[8] "T4 R3 TWP"
[9] "TD R2 WELS"
[10] "PERKINS TOWNSHIP"
```

```
if (nrow(ref_only) > 0) {
  cat("\nSample ref-only districts:\n")
  print(head(ref_only$dist, 10))
}
```

Sample ref-only districts:

[1] "T12/R13 WELS/T9 R8 WELS"	"E TWP"
[3] "TD R2 WELS"	"T11 R4 WELS TWP"
[5] "T15 R6 TWP"	"BIG TWENTY TWP/T15 R15 WELS"
[7] "T4 R3 TWP"	"MOLUNKUS TWP"
[9] "T17 R3 VAN BUREN COVE TWP"	"RANGELEY/ADAMSTOWN TWP"

Eureka, CA

```
eureka_measure <- read.csv("../data/raw/eureka_rcv_2020.csv") %>%
  filter(Precinct != "Totals") %>%
  select(precinct = Precinct, Yes, No, Total.Ballots.Cast) %>%
  mutate(across(c(2:4), parse_number)) %>%
  mutate(yes_share = Yes / Total.Ballots.Cast)

eureka_pres <- read.csv("../data/raw/humboldt_pres_2020.csv") %>%
  select(svprec, total_votes = TOTVOTE, biden = PRSDEM01, trump = PRSREP01) %>%
  mutate(precinct = sub("_A$", "", svprec)) %>%
  group_by(precinct) %>%
  summarize(across(c(2:4), sum))

cat("RCV precincts:", nrow(eureka_measure), "\n")
```

RCV precincts: 11

```
cat("Pres precincts:", nrow(eureka_pres), "\n")
```

Pres precincts: 80

```
eureka_both <- left_join(eureka_measure, eureka_pres, by = "precinct")
cat("After left_join:", nrow(eureka_both), "\n")
```

After left_join: 11

```
cat("NaN in yes_share:", sum(is.nan(eureka_both$yes_share)), "\n")
```

NaN in yes_share: 3

```
cat("NA in yes_share:", sum(is.na(eureka_both$yes_share)), "\n")
```

NA in yes_share: 3

```
# What rows would the NaN filter drop?
nan_rows <- eureka_both %>% filter(is.nan(yes_share))
cat("Rows with NaN yes_share (currently dropped by filter):\n")
```

Rows with NaN yes_share (currently dropped by filter):

```
print(nan_rows)
```

	precinct	Yes	No	Total.Ballots.Cast	yes_share	total_votes	biden	trump
1	1E-28	0	0	0	NaN	0	0	0
2	3E-4J	0	0	0	NaN	0	0	0
3	4E-4J	0	0	0	NaN	0	0	0

Cities: Bloomington, Boulder, Albany, Minnetonka

```
# These use left_join which keeps all LHS rows - just check for NAs
mn_pres <- read.csv("../data/raw/minnesota_pres_2020.csv") %>%
  rename(precinct = PCTNAME)

bloomington_measure <- read.csv("../data/raw/bloomington_rcv_2020.csv") %>%
  rename(precinct = County..Precinct) %>%
  filter(precinct != "Candidate Totals:") %>%
  mutate(precinct = gsub("Hennepin: ", "", precinct))
bloomington_both <- left_join(bloomington_measure, mn_pres, by = "precinct")
cat("Bloomington: RCV precincts:", nrow(bloomington_measure),
    "| After join:", nrow(bloomington_both),
    "| Unmatched:", sum(is.na(bloomington_both$TOTVOTING)), "\n")
```

Bloomington: RCV precincts: 32 | After join: 32 | Unmatched: 0

```

minnetonka_measure <- read.csv("../data/raw/minnetonka_rcv_2020.csv") %>%
  rename(precinct = County..Precinct) %>%
  filter(precinct != "Candidate Totals:") %>%
  mutate(precinct = gsub("Hennepin: ", "", precinct))
minnetonka_both <- left_join(minnetonka_measure, mn_pres, by = "precinct")
cat("Minnetonka: RCV precincts:", nrow(minnetonka_measure),
    "| After join:", nrow(minnetonka_both),
    "| Unmatched:", sum(is.na(minnetonka_both$TOTVOTING)), "\n")

```

Minnetonka: RCV precincts: 23 | After join: 23 | Unmatched: 0

```

albany <- read.csv("../data/raw/albany_all_results_2020.csv")
albany_m <- albany %>% filter(Contest_title == "Measure BB - City of Albany") %>%
  select(Precinct_name) %>% distinct()
albany_p <- albany %>% filter(Contest_title == "President and Vice President") %>%
  select(Precinct_name) %>% distinct()
cat("Albany: Measure precincts:", nrow(albany_m),
    "| Pres precincts:", nrow(albany_p),
    "| Unmatched:", sum(!albany_m$Precinct_name %in% albany_p$Precinct_name), "\n")

```

Albany: Measure precincts: 3 | Pres precincts: 521 | Unmatched: 0

```

boulder <- read.csv("../data/raw/boulder_all_results_2020.csv")
boulder_r <- boulder %>% filter(Contest.Name == "City of Boulder Ballot Question 2E") %>%
  select(Precinct.Name) %>% distinct()
boulder_p <- boulder %>% filter(Contest.Name == "Presidential Electors") %>%
  select(Precinct.Name) %>% distinct()
cat("Boulder: RCV precincts:", nrow(boulder_r),
    "| Pres precincts:", nrow(boulder_p),
    "| Unmatched:", sum(!boulder_r$Precinct.Name %in% boulder_p$Precinct.Name), "\n")

```

Boulder: RCV precincts: 88 | Pres precincts: 240 | Unmatched: 0

Summary: Final Dataset

```

load("../data/rcv_data.RData")

cat("Total rows:", nrow(states_and_cities), "\n\n")

```

Total rows: 3446

```
cat("NAs per column:\n")
```

NAs per column:

```
print(colSums(is.na(states_and_cities)))
```

district	2866
ak_rcv_race	2867
ak_rcv_yes	2867
ak_rcv_no	2867
ak_pres_race	2866
ak_pres_dem	2866
ak_pres_con	2866
ak_pres.ali	2866
ak_pres.grn	2866
ak_pres.lib	2866
ak_pres.nom	2866
ak_pres.rep	2866
rcv_yes	6
rcv_no	6
yes_share	10
dem_share	8
precinct_id	0
locale	0
state	0
city	3289
rcv_jurisdiction	0
Locality.x	1273
Ward.x	1273
Pct.x	1273
pres_biden	1182
pres_trump	650
pres_jorg	1273

	pres_hawk	
	1273	
	pres_other	
	1273	
	pres_nopref	
	1273	
	pres_blank	
	741	
	pres_total	
	639	
	precinct	
	1119	
	biden_share	
	1182	
	Locality.y	
	1273	
	Ward.y	
	1273	
	Pct.y	
	1273	
	rcv_blank	
	742	
	rcv_total	
	1171	
	county_code	
	2910	
	dist	
	2910	
	pres_clinton	
	2914	
	pres_johnson	
	2914	
	pres_stein	
	2914	
	pres_castle	
	2914	
	pres_fox	
	2914	
	pres_kotlikoff	
	2914	
	pres_mcmullin	
	2914	
	clinton_share	
	2914	
	X.1	
	2915	
Question.1..Citizen.Initiative		
	2915	
	X.3	
	2915	
	X.4	
	2915	
Question.2..Citizen.Initiative		
	2915	

	X.5
	2915
	X.6
	2915
Question.3..Citizen.Initiative	
	2915
	X.7
	2915
	X.8
	2915
Question.4..Citizen.Initiative	
	2915
	X.9
	2915
	X.10
	2915
Question.6...Bond.Issue	
	2915
	X.13
	2915
	X.14
	2915
maine_total_ballots	
	2915
	X.15
	3446
	X.16
	3446
	X.17
	3446
	X.18
	3446
	X.19
	3446
	X.20
	3446
	X.21
	3446
	X.22
	3446
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	2915
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Reporting_flag	
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	3443
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Pct_seq_nbr	
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Reg_voters	3443
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```

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USREPLMN	
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	3391
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USREPTOTAL	
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	3358
	pres_Bill Hammons / Eric Bodenstab
	3358
	pres_Howie Hawkins / Angela Nicole Walker
	3358
	pres_Blake Huber / Frank Atwood
	3358
	pres_Jo Jorgensen / Jeremy "Spike" Cohen
	3358
	pres_Brian Carroll / Amar Patel
	3358
	pres_Mark Charles / Adrian Wallace
	3358
	pres_Phil Collins / Billy Joe Parker
	3358
	pres_Roque "Rocky" De La Fuente / Darcy G. Richardson
	3358
	pres_Dario Hunter / Dawn Neptune Adams
	3358
	pres_Princess Khadijah Maryam Jacob-Fambro / Khadijah Maryam Jacob Sr.
	3358
	pres_Alyson Kennedy / Malcolm Jarrett
	3358
	pres_Joseph Kishore / Norissa Santa Cruz
	3358
	pres_Kyle Kenley Kopitke / Nathan Re Vo Sorenson
	3358
	pres_Gloria La Riva / Sunil Freeman
	3358
	pres_Joe McHugh / Elizabeth Storm
	3358

pres_Brock Pierce / Karla Ballard	3358
pres_Jordan "Cancer" Scott / Jennifer Tepool	3358
pres_Kanye West / Michelle Tidball	3358
pres_Kasey Wells / Rachel Wells (Write-in)	3358
pres_Todd Cella / Timothy Bryan Cella (Write-in)	3358
pres_Tom Hoefling / Andy Prior (Write-in)	3358
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3435
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3435
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3435
recent_lpw
0
```

```
cat("\nNAs by place:\n")
```

NAs by place:

```
states_and_cities %>%
  group_by(locale) %>%
  summarize(
    n = n(),
    na_precinct_id = sum(is.na(precinct_id)),
    na_dem_share = sum(is.na(dem_share)),
    na_yes_share = sum(is.na(yes_share)),
    .groups = "drop"
  ) %>%
  print(n = 10)
```

A tibble: 8 x 5

	locale	n	na_precinct_id	na_dem_share	na_yes_share
	<chr>	<int>	<int>	<int>	<int>
1	Alaska	580	0	1	2
2	Albany	3	0	0	0
3	Bloomington	32	0	0	0
4	Boulder	88	0	0	0
5	Eureka	11	0	3	3
6	Maine	536	0	4	5
7	Massachusetts	2173	0	0	0
8	Minnetonka	23	0	0	0

```
cat("\nAll rows with any NA:\n")
```

All rows with any NA:

```
states_and_cities %>%
  filter(if_any(everything(), is.na)) %>%
  print(n = 30)
```

A tibble: 3,446 x 328

	district	ak_rcv_race	ak_rcv_yes	ak_rcv_no	ak_pres_race	ak_pres_dem
	<chr>	<chr>	<int>	<int>	<chr>	<int>
1	"01-446 Aurora "	Ballot Mea~	329	410	U.S. Presid~	239
2	"01-455 Fairbanks ~	Ballot Mea~	58	80	U.S. Presid~	47

3	"01-465 Fairbanks ~ Ballot Mea~	112	103 U.S. Presid~	85
4	"01-470 Fairbanks ~ Ballot Mea~	161	186 U.S. Presid~	128
5	"01-475 Fairbanks ~ Ballot Mea~	109	113 U.S. Presid~	103
6	"01-480 Fairbanks ~ Ballot Mea~	252	356 U.S. Presid~	209
7	"01-485 Fairbanks ~ Ballot Mea~	165	177 U.S. Presid~	140
8	"01-490 Fairbanks ~ Ballot Mea~	196	231 U.S. Presid~	165
9	"01-495 Fairbanks ~ Ballot Mea~	81	89 U.S. Presid~	54
10	"02-345 Badger No.~ Ballot Mea~	370	652 U.S. Presid~	183
11	"02-355 Fairbanks ~ Ballot Mea~	222	334 U.S. Presid~	181
12	"02-365 Fairbanks ~ Ballot Mea~	364	462 U.S. Presid~	236
13	"02-375 Fort Wainw~ Ballot Mea~	140	146 U.S. Presid~	74
14	"03-130 Badger No.~ Ballot Mea~	349	750 U.S. Presid~	179
15	"03-135 Chena Lake~ Ballot Mea~	304	723 U.S. Presid~	130
16	"03-165 Newby " Ballot Mea~	356	837 U.S. Presid~	160
17	"03-175 North Pole~ Ballot Mea~	190	362 U.S. Presid~	98
18	"03-183 Plack " Ballot Mea~	291	723 U.S. Presid~	116
19	"04-230 Ester " Ballot Mea~	236	152 U.S. Presid~	204
20	"04-240 Farmers Lo~ Ballot Mea~	368	400 U.S. Presid~	306
21	"04-250 Goldstream~ Ballot Mea~	263	204 U.S. Presid~	220
22	"04-260 Goldstream~ Ballot Mea~	348	234 U.S. Presid~	315
23	"04-265 Steese Eas~ Ballot Mea~	367	760 U.S. Presid~	228
24	"04-270 Steese Wes~ Ballot Mea~	428	732 U.S. Presid~	266
25	"04-280 University~ Ballot Mea~	82	63 U.S. Presid~	71
26	"05-580 Airport " Ballot Mea~	38	73 U.S. Presid~	22
27	"05-582 Chena " Ballot Mea~	598	783 U.S. Presid~	452
28	"05-586 Geist " Ballot Mea~	267	281 U.S. Presid~	202
29	"05-587 Lakeview " Ballot Mea~	40	87 U.S. Presid~	27
30	"05-588 Pike " Ballot Mea~	89	143 U.S. Presid~	66

i 3,416 more rows

i 322 more variables: ak_pres_con <int>, ak_pres.ali <int>,
ak_pres.grn <int>, ak_pres.lib <int>, ak_pres.nom <int>, ak_pres.rep <int>,
rcv.yes <dbl>, rcv.no <dbl>, yes.share <dbl>, dem.share <dbl>,
precinct_id <chr>, locale <chr>, state <chr>, city <chr>,
rcv.jurisdiction <chr>, Locality.x <chr>, Ward.x <chr>, Pct.x <chr>,
pres.biden <dbl>, pres.trump <dbl>, pres.jorg <dbl>, pres.hawk <dbl>, ...